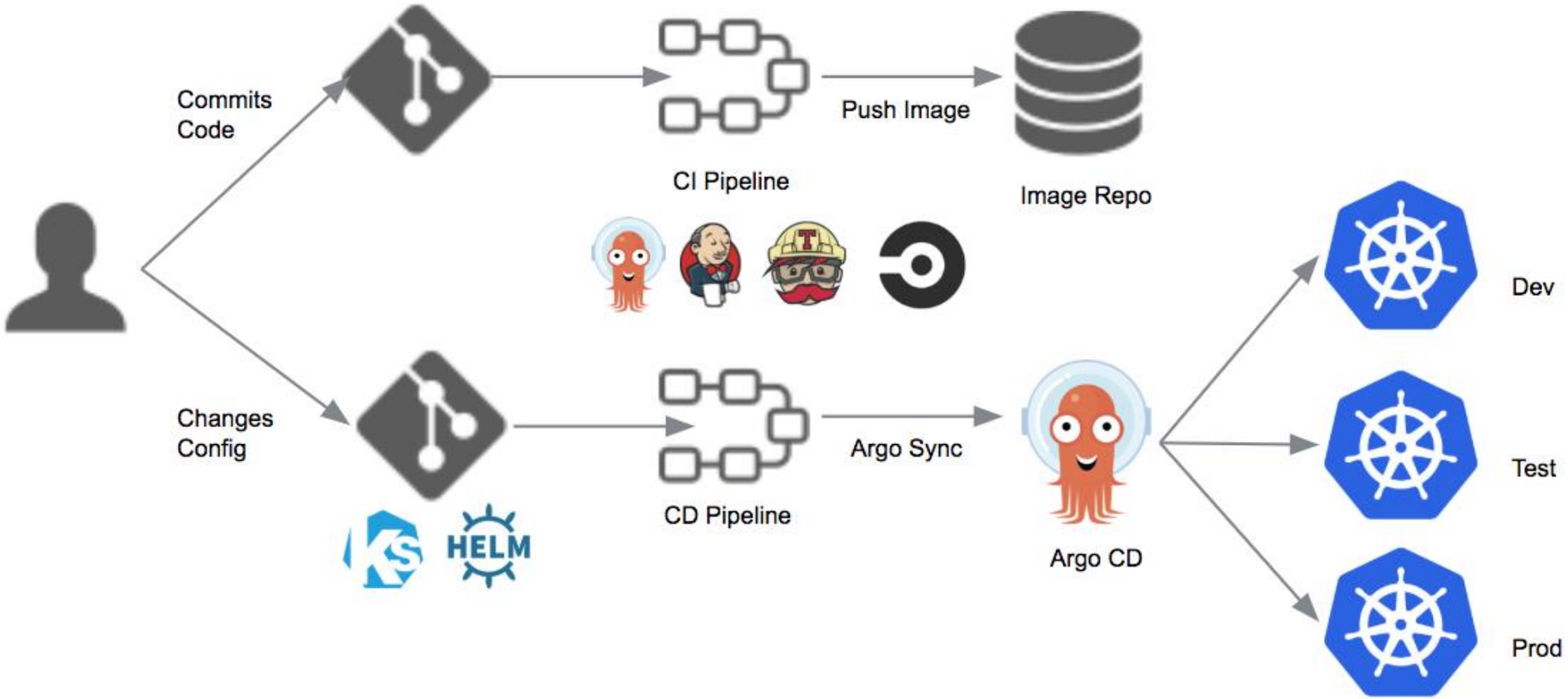
Developer Operations (devops)

Adaptions to Deployment

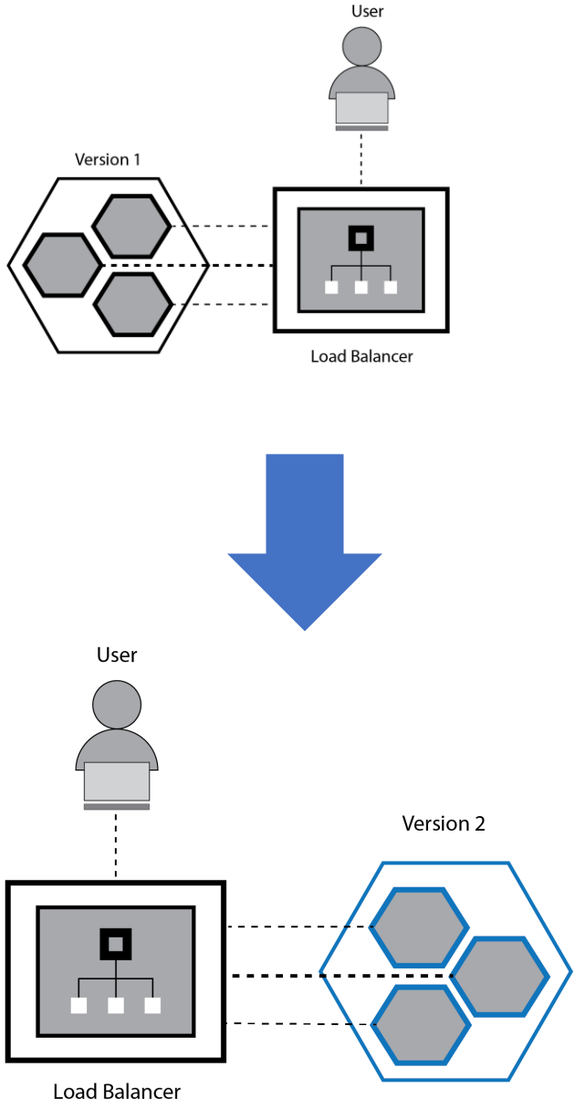
Recap: How to deploy?



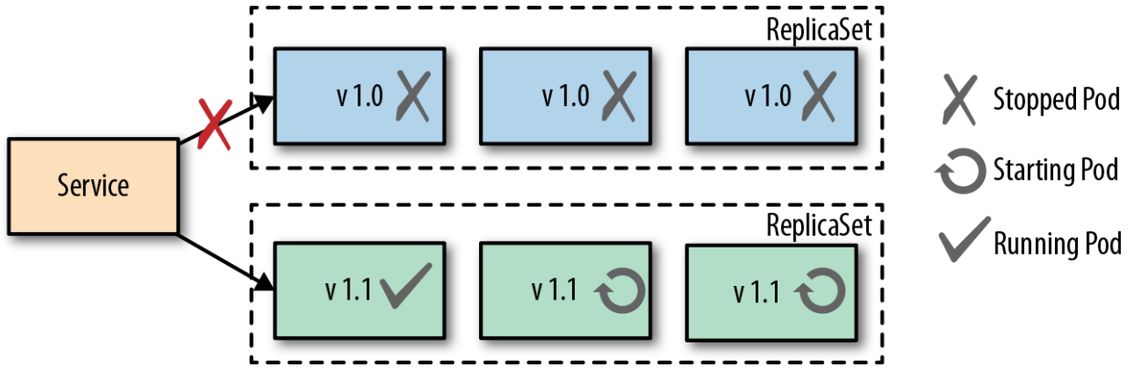
How do you rollout?

Recreate Deployment

Removes old versioned pods by replacing them: all at one time



Recreate Deployment, Implementation




```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: random-generator
spec:
  replicas: 3
  strategy:
    type: Recreate
    selector:
      matchLabels:
        app: random-generator
    template:
      metadata:
        labels:
          app: random-generator
      spec:
        containers:
        - image: k8spatterns/random-generator:1.0
          name: random-generator
          readinessProbe:
            exec:
              command: [ "stat", "/random-generator-ready" ]
```

Recreate Deployment

Removes old versioned pods one by one by replacing it.

Pros:

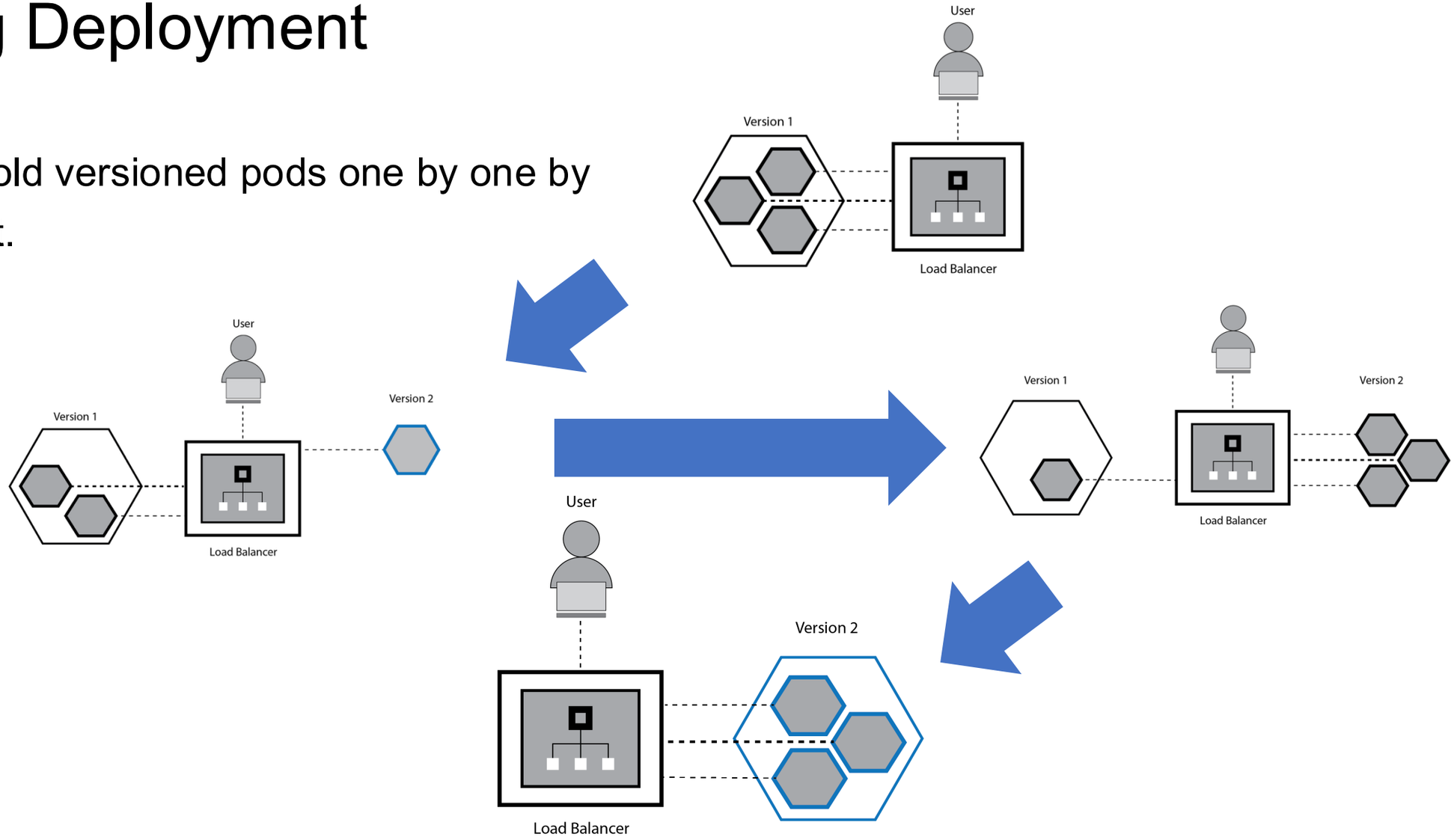
- Einfach
- Nur eine Version erreichbar, Breaking Changes
- Steuerung exakter Zeitpunkt
- ApplikationState, auch extern komplett deployt

Cons:

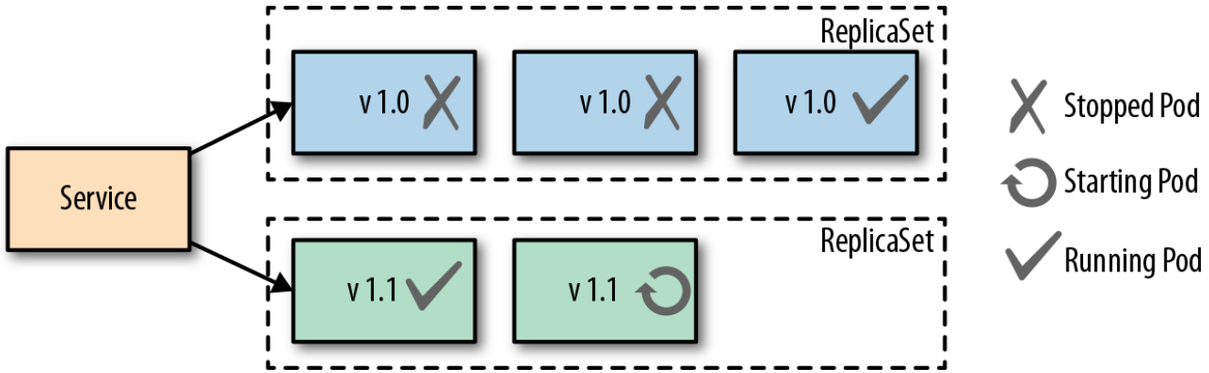
- Downtime

Rolling Deployment

Removes old versioned pods one by one by replacing it.



Rolling Deployment, Implementation




```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: random-generator
spec:
  replicas: 3
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 1
    selector:
      matchLabels:
        app: random-generator
    template:
      metadata:
        labels:
          app: random-generator
      spec:
        containers:
        - image: k8spatterns/random-generator:1.0
          name: random-generator
          readinessProbe:
            exec:
              command: [ "stat", "/random-generator-ready" ]
```

Rolling Deployment

Removes old versioned pods one by one by replacing it.

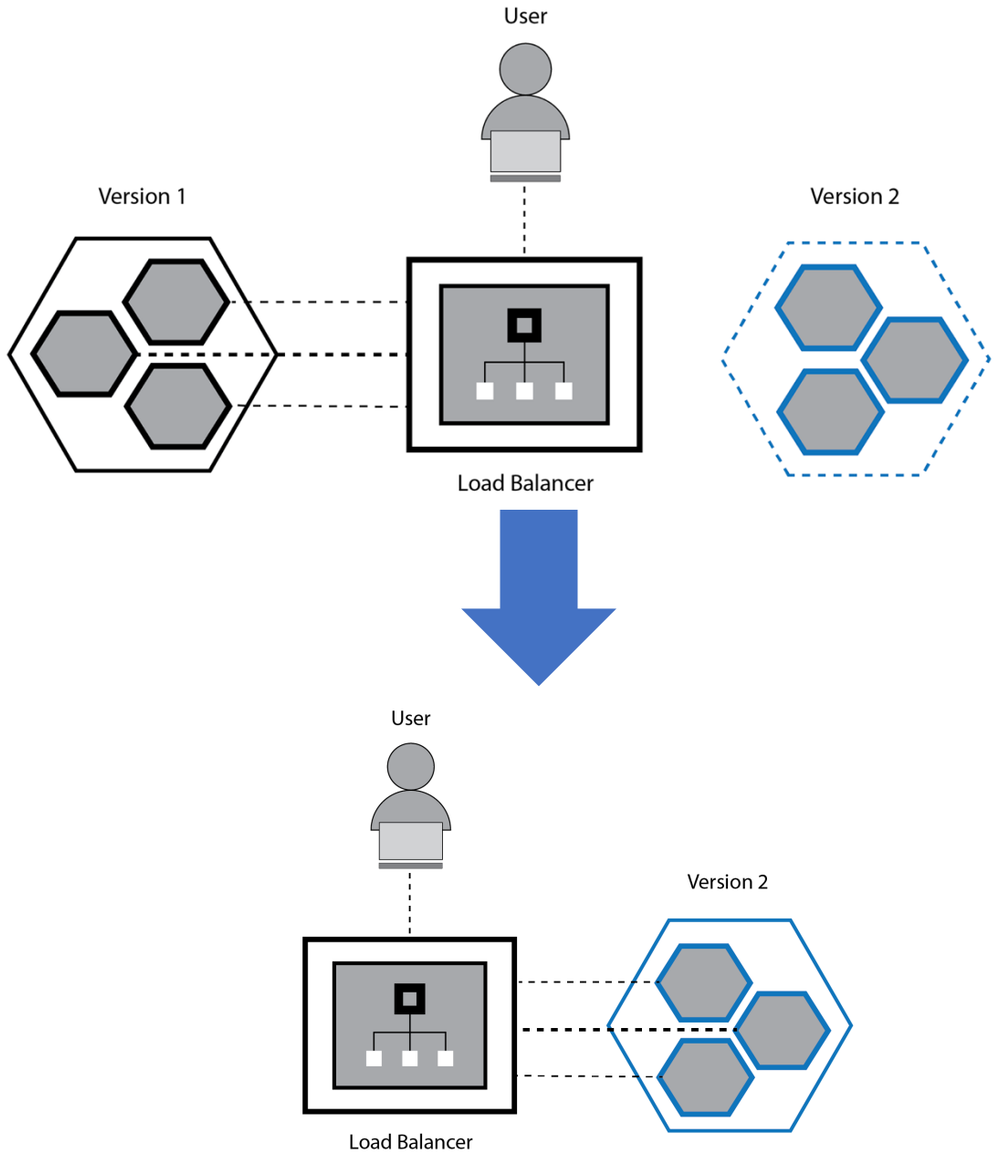
Pros:

- Keine Downtime
- Rollback bei der Applikation
- Easy to implement

Cons:

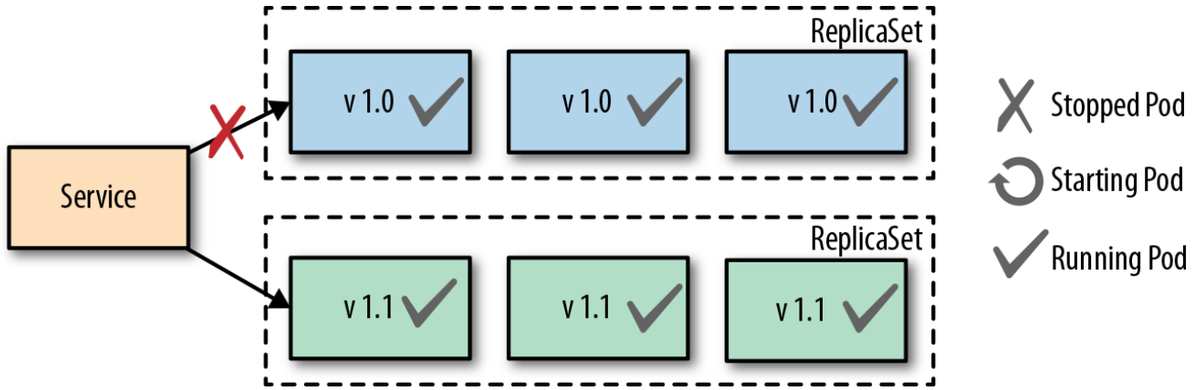
- Seiteneffekt / Umsysteme
- “Unkontrolliert”, Zeit wie auch Traffic

Blue/Green Deployment

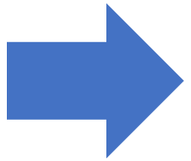


Duplicate deployment, new version is deployed aside to old one. Switching traffic to new version.

Blue/Green Deployment




```
apiVersion: v1
kind: Service
metadata:
  name: service
spec:
  selector:
    app: myapp
    env: blue
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
```



```
apiVersion: v1
kind: Service
metadata:
  name: service
spec:
  selector:
    app: myapp
    env: green
  ports:
  - protocol: TCP
    port: 80
    targetPort: 80
```

Blue/Green Deployment

Duplicate deployment, new version is deployed aside to old one. Switching traffic to new version.

Pros:

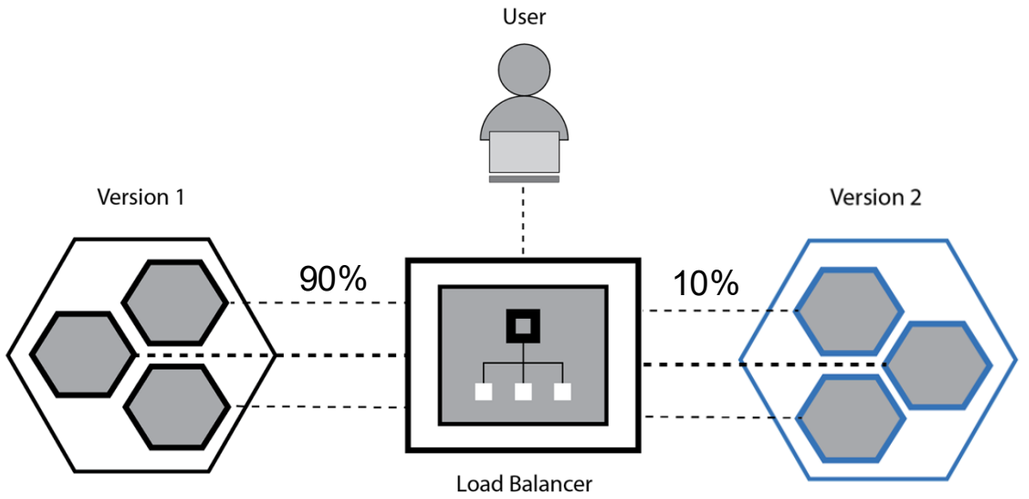
- Zeitliche Kontrolle, atomar
- Rollbackfähigkeit (beware Umsysteme)
- No Downtome

Cons:

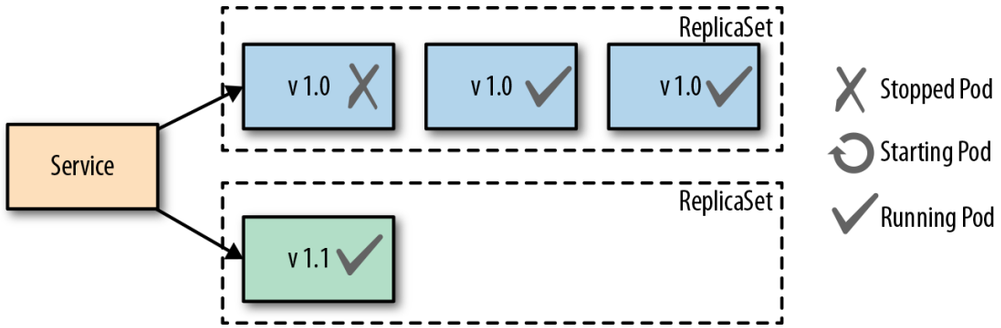
- Doppelte Ressource

Canary Deployment

Redirecting small portions of traffic to new release.



Canary Deployment, Implementation




```yaml
apiVersion: v1
kind: Service
metadata:
  name: service
spec:
  selector:
    app: random-generator
ports:
  - protocol: TCP
    port: 80
    targetPort: 80
```

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: random-generator-1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: random-generator
  template:
    metadata:
      labels:
        app: random-generator
    spec:
      containers:
        - image: k8spatterns/random-generator:1.0
          name: random-generator
          readinessProbe:
            exec:
              command: [ "stat", "/random-generator-ready" ]
```

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: random-generator-2
spec:
  replicas: 1
  selector:
    matchLabels:
      app: random-generator
  template:
    metadata:
      labels:
        app: random-generator
    spec:
      containers:
        - image: k8spatterns/random-generator:2.0
          name: random-generator
          readinessProbe:
            exec:
              command: [ "stat", "/random-generator-ready" ]
```

Canary Deployment

Redirecting small portions of traffic to new release.

Pros:

- Impact auf Userbasis wird minimiert
- Kontrolle / zeitliche Planbarkeit
- Keine signifikanten Mehrkosten

Cons:

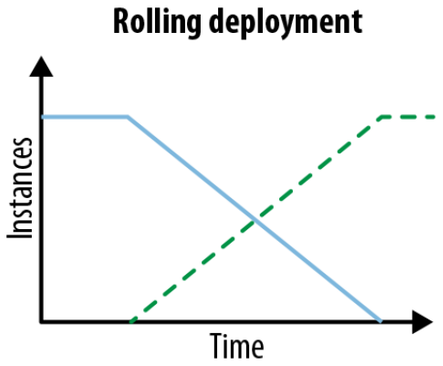
- Multi Api Support
- Complicated to implement, other tools become necessary such as service mesh

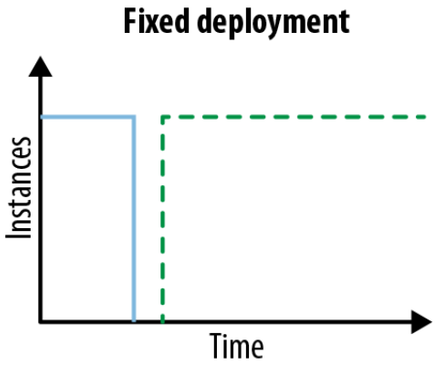
Comparison, Deployment Patterns

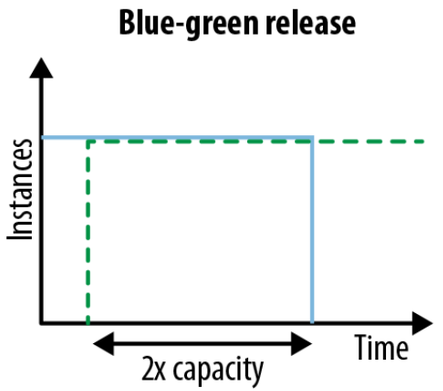
Different Patterns for different use cases:

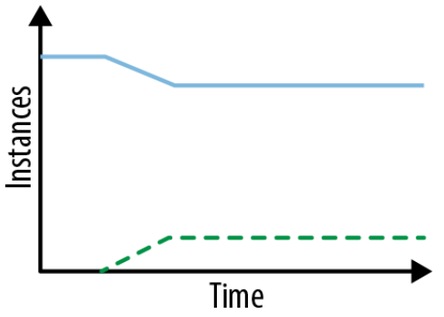
- Utilized external systems like databases?
- Costs for duplicate infrastructure?
- Rollback ability regarding the application?
- Additional traffic redirection techniques like service mesh available?

There is no one-size-fits-all…









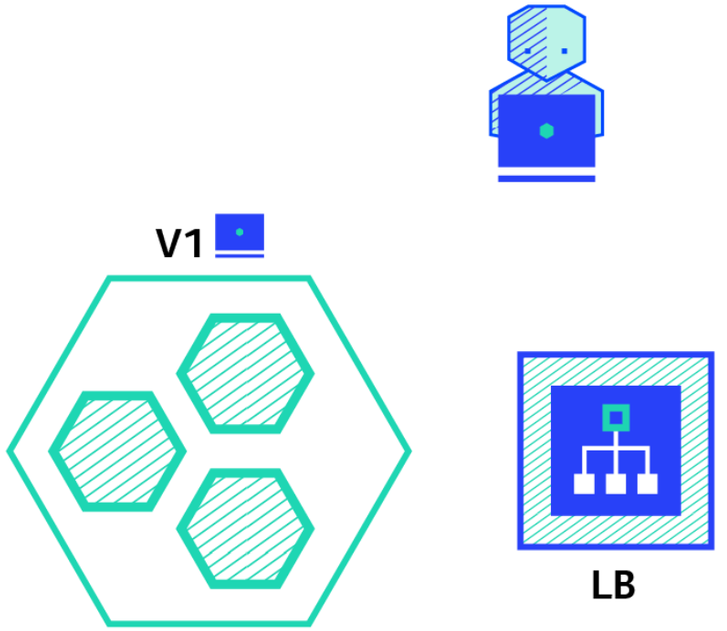
Additional Deployment Strategies, A/B Testing

Traffic for a given part of users is redirected

+ Control about traffic

+ Opt-in from user side

- Complex setup, semantic load balancing
- Complex to trace errors, especially in backing services



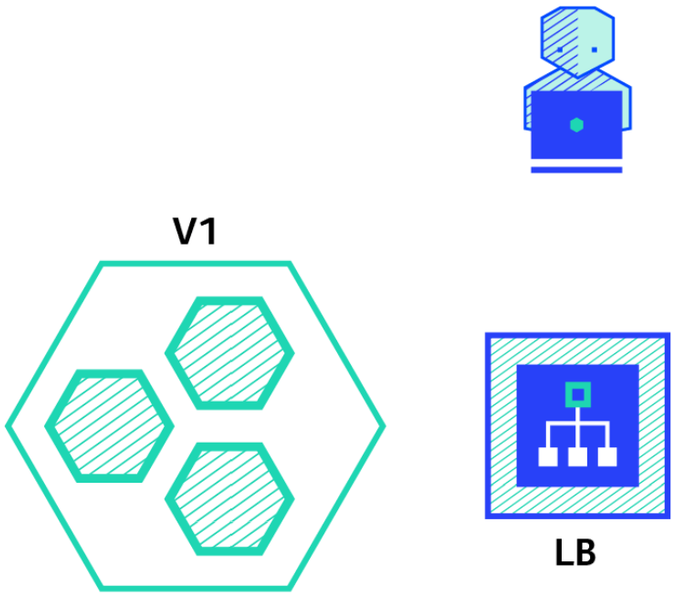
Deployment Strategies, Shadowing

Requests are send to new release as well, acting as a shadow

+ no direct user impact

+ load testing without load testing

+ switch when everything is ready

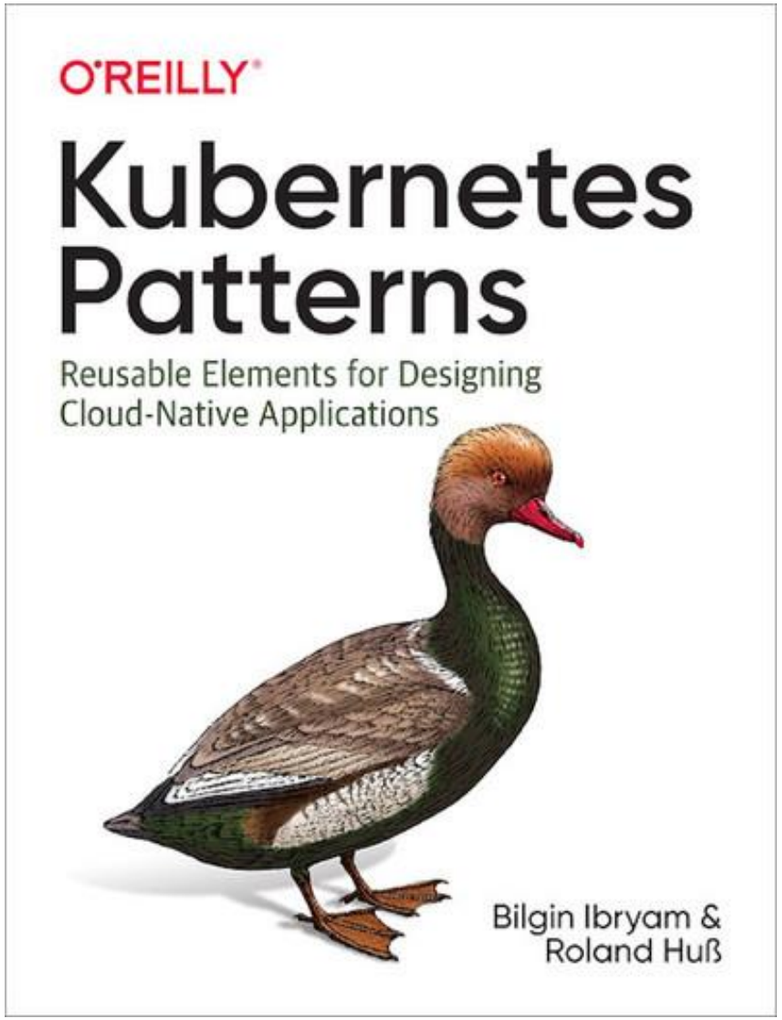


- Double infrastructure
- Entire shadowing can become complex with respect to backing services
- Complex to set up

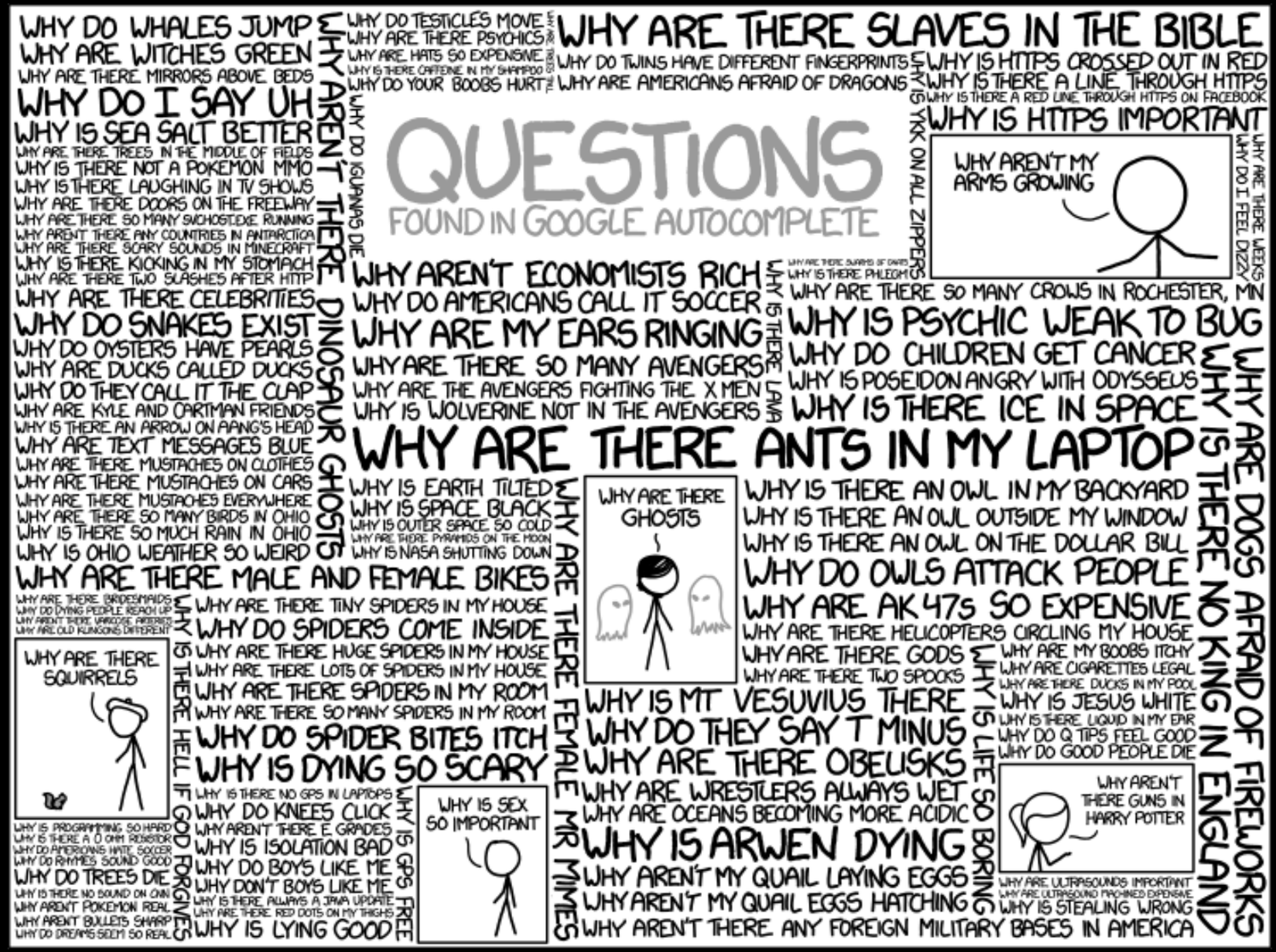
Deployment Strategies, Summary

Strategy	ZERO DOWNTIME	REAL TRAFFIC TESTING	TARGETED USERS	CLOUD COST	ROLLBACK DURATION	NEGATIVE IMPACT ON USER	COMPLEXITY OF SETUP
RECREATE version A is terminated then version B is rolled out	✗	✗	✗	■□□	■■■	■■■	□□□
RAMPED version B is slowly rolled out and replacing version A	✓	✗	✗	■□□	■■■	■□□	■□□
BLUE/GREEN version B is released alongside version A, then the traffic is switched to version B	✓	✗	✗	■■■	□□□	■■□	■■□
CANARY version B is released to a subset of users, then proceed to a full rollout	✓	✓	✗	■□□	■□□	■□□	■■□
A/B TESTING version B is released to a subset of users under specific condition	✓	✓	✓	■□□	■□□	■□□	■■■
SHADOW version B receives real world traffic alongside version A and doesn't impact the response	✓	✓	✗	■■■	□□□	□□□	■■■

Literature


O'REILLY
Kubernetes Patterns
Reusable Elements for Designing Cloud-Native Applications
Bilgin Ibryam & Roland Huß


Onur Yilmaz
Kubernetes Design Patterns and Extensions
Enhance your container-cluster management skills and efficiently develop and deploy applications
Packt


QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SYCHOSTICKS RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REARM UP
WHY AREN'T THERE VARICOSE AIRBIKES
WHY ARE OLD KLINGONS DIFFERENT
WHY ARE THERE SQUIRRELS
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO SOUND ON CNN
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY ARE THERE SLAVES IN THE BIBLE
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY IS YKK ON ALL ZIPPERS
WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY
WHY DO IGUANAS DIE
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY ARE THERE SWARMS OF GNATS
WHY IS THERE PHLEGM
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY IS THERE LAVA
WHY ARE THERE ANTS IN MY LAPTOP
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE GHOSTS
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY IS SEX SO IMPORTANT
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND MACHINES EXPENSIVE
WHY IS STEALING WRONG
WHY ARE DOGS AFRAID OF FIREWORKS
WHY IS THERE NO KING IN ENGLAND
WHY IS LIFE SO BORING
WHY IS GPS FREE
WHY ARE THERE FEMALE MR MIMES
WHY IS THERE HELL IF GOD FORGIVES
WHY AREN'T THERE DINOSAUR GHOSTS